AQA GCSE Biology – Organisation

Revision Notes

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1. Organisation in Living Organisms

Living organisms are highly organised. In multicellular organisms such as humans, cells are specialised and work together to carry out complex functions. Organisation allows the body to operate efficiently and maintain life processes.

The levels of organisation in living organisms follow a clear hierarchy:

Cells → Tissues → Organs → Organ Systems → Organism

- **Cells** are the smallest units of life.
- **Tissues** are groups of similar cells working together.
- **Organs** are structures made of different tissues performing specific functions.
- **Organ systems** are groups of organs working together to perform major body functions.
- **Organisms** are the complete living individual.

Example in humans:

- **Muscle cells** form **muscle tissue**
- Muscle tissue forms the **heart**
- The heart works within the **circulatory system**

This hierarchy ensures that specialised structures can perform particular functions efficiently.

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2. Animal Tissues

Animals contain several different tissue types, each adapted for a specific function.

Muscle Tissue

Muscle tissue contracts to produce movement. There are three main types:

1. **Skeletal muscle** – attached to bones and responsible for voluntary movement.
2. **Cardiac muscle** – found only in the heart and contracts rhythmically.

3. **Smooth muscle** – found in internal organs such as the digestive system.

Muscle cells contain specialised protein fibres that shorten during contraction.

Glandular Tissue

Glandular tissue is responsible for producing and releasing substances in the body.

Examples include:

- **Salivary glands** producing saliva
- **Sweat glands** producing sweat
- **Endocrine glands** producing hormones

These substances are secreted either into ducts or directly into the bloodstream.

Epithelial Tissue

Epithelial tissue covers the surfaces of organs and lines body cavities.

Functions include:

- Protection
- Absorption
- Secretion
- Filtration

Examples include:

- Skin surface
- Lining of the digestive system
- Inner lining of blood vessels

Epithelial cells are tightly packed together to form protective layers.

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3. Animal Organs

An organ is a structure made up of different tissues working together to perform a specific function.

Examples of human organs include:

- Heart
- Stomach
- Lungs
- Liver
- Brain

Each organ contains multiple tissue types.

Example: **The stomach**

The stomach contains:

- **Muscle tissue** – churns food
- **Glandular tissue** – produces digestive enzymes and acid
- **Epithelial tissue** – lines the stomach and protects against acid

Together, these tissues allow the stomach to digest food efficiently.

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4. The Human Digestive System

The digestive system breaks down food into small soluble molecules that can be absorbed into the bloodstream.

The main organs involved are:

- Mouth
- Oesophagus
- Stomach
- Small intestine
- Large intestine
- Liver
- Pancreas
- Gall bladder

The digestive system performs four main processes:

1. **Ingestion** – taking food into the body.
2. **Digestion** – breaking food into smaller molecules.
3. **Absorption** – movement of nutrients into the blood.
4. **Egestion** – removal of undigested waste.

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5. Mechanical and Chemical Digestion

Mechanical Digestion

Mechanical digestion physically breaks food into smaller pieces without changing its chemical composition.

Examples include:

- **Chewing in the mouth**
- **Churning in the stomach**

This increases the surface area of food so enzymes can work more effectively.

Chemical Digestion

Chemical digestion involves enzymes breaking down large insoluble molecules into small soluble molecules.

Examples:

- Starch → Glucose
- Proteins → Amino acids
- Lipids → Fatty acids and glycerol

Enzymes are essential for these reactions.

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6. Digestive Enzymes

Digestive enzymes are biological catalysts that speed up the breakdown of food molecules.

There are three main groups of digestive enzymes:

Carbohydrases

Carbohydrases break down carbohydrates.

Example:

Amylase breaks down starch into sugars.

Produced in:

- Salivary glands
- Pancreas
- Small intestine

Proteases

Proteases break down proteins into amino acids.

Produced in:

- Stomach (pepsin)
- Pancreas
- Small intestine

Protein digestion begins in the stomach.

Lipases

Lipases break down lipids (fats) into fatty acids and glycerol.

Produced mainly by the **pancreas** and released into the **small intestine**.

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7. The Role of Bile

Bile is a digestive substance produced by the **liver** and stored in the **gall bladder**.

Bile has two important functions:

1. Neutralising Stomach Acid

Food entering the small intestine from the stomach is acidic due to hydrochloric acid.

Bile is **alkaline**, so it neutralises the acid and creates the ideal pH for enzymes in the small intestine.

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2. Emulsifying Fats

Bile breaks large fat droplets into smaller droplets.

This process is called **emulsification**.

Emulsification increases the surface area of fats, allowing **lipase enzymes** to digest fats more quickly.

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8. The Small Intestine and Absorption

Most digestion and absorption occur in the **small intestine**.

The inner surface of the small intestine contains millions of tiny finger-like projections called **villi**.

These increase the surface area available for absorption.

Adaptations of Villi

Villi are highly specialised for efficient absorption.

Key adaptations include:

- **Large surface area** for absorption
- **Thin walls** (one cell thick) for fast diffusion
- **Rich blood supply** to carry absorbed nutrients away
- **Microvilli** on the surface of cells to further increase surface area

Nutrients absorbed include:

- Glucose
- Amino acids
- Fatty acids
- Glycerol
- Vitamins
- Minerals

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9. The Large Intestine

The large intestine absorbs water from undigested food material.

Functions include:

- Reabsorbing water
- Forming faeces
- Housing beneficial bacteria

The remaining waste is stored in the **rectum** before being removed through the **anus**.

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10. The Circulatory System

The circulatory system transports substances around the body.

It consists of three main components:

1. **Heart** – pumps blood
2. **Blood vessels** – carry blood
3. **Blood** – transports substances

The circulatory system transports:

- Oxygen
- Carbon dioxide
- Nutrients
- Hormones
- Waste products

Humans have a **double circulatory system**.

This means blood passes through the heart twice in each complete circulation.

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11. Blood Vessels

There are three main types of blood vessels.

Arteries

Arteries carry blood **away from the heart**.

Features:

- Thick muscular walls
- Small lumen
- High pressure blood flow

Example: **Aorta**

Veins

Veins carry blood **towards the heart**.

Features:

- Thin walls
- Large lumen
- Contain **valves** to prevent backflow

Capillaries

Capillaries are very small vessels where exchange of substances occurs.

Features:

- Walls only **one cell thick**
- Very narrow lumen
- Large surface area

This allows efficient diffusion of oxygen, nutrients, and waste products.

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12. The Heart

The heart is a muscular organ that pumps blood around the body.

It has four chambers:

- **Right atrium**
- **Right ventricle**
- **Left atrium**

- **Left ventricle**

The **left ventricle** has the thickest wall because it pumps blood to the entire body.

Blood flow through the heart:

Body → Right atrium → Right ventricle → Lungs → Left atrium → Left ventricle → Body

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13. Blood Components

Blood consists of several components.

Red Blood Cells

Function: transport oxygen.

Adaptations:

- Contain **haemoglobin**
- Biconcave shape increases surface area
- No nucleus, allowing more space for haemoglobin

White Blood Cells

Function: defend the body against pathogens.

They work by:

- Engulfing pathogens
- Producing antibodies
- Producing antitoxins

Platelets

Platelets are small cell fragments involved in **blood clotting**.

They prevent excessive blood loss and help protect against infection.

Plasma

Plasma is the liquid part of blood.

It transports:

- Carbon dioxide
- Urea
- Hormones
- Nutrients
- Heat

Plasma makes up about **55% of blood volume**.

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14. Coronary Heart Disease

Coronary heart disease occurs when the **coronary arteries** become narrowed due to fatty deposits called **atheroma**.

This reduces blood flow to the heart muscle.

Possible consequences include:

- Angina (chest pain)
- Heart attack

Risk factors include:

- Smoking
- High-fat diet
- Lack of exercise
- Obesity
- High blood pressure

Treatments include:

- Lifestyle changes
- Statins
- Stents
- Bypass surgery

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15. Lifestyle and Cardiovascular Health

Lifestyle choices strongly influence cardiovascular health.

Healthy habits include:

- Regular exercise
- Balanced diet
- Not smoking
- Maintaining a healthy weight

Exercise strengthens the heart and improves circulation.

A balanced diet helps prevent the buildup of fatty deposits in arteries.

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16. Health and Disease

Health is defined as **the state of physical and mental well-being**.

Diseases can be:

- **Communicable** (infectious)
- **Non-communicable**

Factors affecting health include:

- Diet
- Stress
- Lifestyle
- Genetics
- Environment

Maintaining a healthy lifestyle reduces the risk of many diseases.